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INTRODUCTION TO PHYSICS-INSPIRED ENERGY TRANSITION NEURAL NETWORK (PETNN)



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Ye Kyaw Thu earned his B.Sc. in Physics from Dagon University, Myanmar, in 2000, followed by an M.Sc. (2006) and D.Sc. (2011) in Global Information and Telecommunication Studies from Waseda University, Japan. His research focuses on artificial intelligence, natural language processing (NLP), and human-computer interaction (HCI). He founded the Language Understanding Lab in Myanmar and has been a Visiting Professor with the Language & Semantic Technology Research Team (LST) at the Artificial Intelligence Research Unit (AINRU), National Electronics and Computer Technology Center (NECTEC), Thailand, since 2019. He actively supervises and co-supervises undergraduate, master's, and doctoral students at several institutions, including Assumption University (AU), King Mongkut's Institute of Technology Ladkrabang (KMUTL), Kasetsart University (KU), and Sirindhorn International Institute of Technology (SIIT).

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Abstract

On May 6, 2025, a preprint paper introduced the Physics-inspired Energy Transition Neural Network (PETNN), a novel recurrent neural network (RNN) architecture inspired by physics-based energy transition models for sequence learning (Wu et al., 2025). The authors demonstrated PETNN's efficacy in tasks like MNIST classification and sentiment analysis, addressing long-term dependencies. Inspired by their work, I replicated their experiments and developed two new architectures for Myanmar language datasets, despite the absence of public source code. In this lecture, I will present state-of-the-art (SOTA) results using PETNN for digit classification on Myanmar MNIST, a novel Siamese PETNN for paraphrase detection with myParaphrase, and the first PETNN-based text generation with myPOS. These advancements showcase PETNN's versatility for low-resource language processing.